Gary Chang

+1-903-907-2995 | gxc180016@utdallas.edu | gchang1020.github.io

Data and geospatial scientist with two decades of software/systems engineering experience, applying Bayesian, causal-inference, and network methods to build data pipelines and models from messy, real-world data.

EXPERIENCE

Research / Teaching Assistant; University of Texas at Dallas; 9/2020 – Present; Dallas

- Conducted applied research across political science, public health, GIS suitability analysis, remote sensing, and operations research, applying classical statistics, machine learning, network analysis (QAP/ERGM), and Bayesian/INLA methods.
- Taught undergraduate courses and supported graduate-level instruction in quantitative methods and data science, including course design, student mentorship, and evaluation.

Senior Software Engineer; WheelsAmerica; 8/2018 – 5/2021; Dallas

- Designed and built codegen tooling for data mappings and RESTful interfaces (PHP, Phalcon, AWS, WordPress, Google Maps, Cloudinary), connecting the company's inventory and accounting systems with eBay, Amazon, and SureDone.

Marketing Advisor; VSO/UN on Tanzania Local Enterprise Development (T-LED); 3/2016 – 7/2018; Mwanza, Tanzania

- Supported SME growth across Malawi, Rwanda, Uganda, and Tanzania through market research, training delivery, business/marketing plans, fundraising proposals, and websites.

Foreign Scholar; Peking University, School of Software and Microelectronics; 9/2013 – 2/2016; Beijing, China

- Lectured graduate students on professional and corporate skills (debate, proposal writing, strategic marketing, presentation).
- Led the "Classroom Automation" project (Linux/Apache/MySQL/PHP) for the National Challenge Cup Tournament, including mobile submission, auto-grading, and resource sharing.

Additional Professional Experience, 1996 – 2013

Senior Director of Globalization, American Express, New York (2010–2013): Led global deployment of the SERVE online payment platform across legal, compliance, I18N/L10N, and IT infrastructure workstreams.

VP of Engineering, Xubo Media Ltd, Hong Kong (2006–2010): Built the ABITO mobile ad platform and the AdEstate cross-channel ad trading system; led CRM/SCM systems integration.

Project Officer, Plateau Homeland Conservation, Tibet, China (2005–2006): Coordinated fundraising and resources for livestock infrastructure projects.

IT Partner, Mercatela Ltd, Tokyo, Japan (2003–2005): Built Foodbiz.com for Nikkei Business Publications (J2EE, Hibernate, Struts, Oracle DB2, WebSphere).

Principal Consultant, IBM Inc., Burlingame, CA (2000–2003): CRM/SCM requirements and EAI architecture for Manugistics, BetzDearborn, and US West.

Sr. Software Engineer, SAP Technology Inc., Palo Alto, CA (1996–2000): Led Sales Force Automation design team; built RFC Class Library and CORBA prototypes.

RESEARCH & APPLIED PROJECTS

- Doctoral research (UT Dallas): built an LLM-assisted data extraction pipeline and 280+ observation panel dataset; applied survival analysis (Cox/frailty), GLM/GLMM, network analysis (QAP/ERGM), and Bayesian/INLA to model institutional elite dynamics in a closed political system.
- NLP and geospatial analytics: geocoded Twitter sentiment analysis of the 2022 Qatar World Cup using a RoBERTa transformer model with Nominatim/OpenStreetMap geocoding; YOLO/RoboFlow-based drone object detection for GIS suitability analysis.
- Optimization and predictive modeling: flight sequence risk mitigation combining XGBoost gradient boosting with Google OR-Tools integer programming; spatial-temporal Bayesian/INLA modeling of renal disease mortality (Texas); ArcGIS Pro suitability modeling and multi-criteria decision making (MCDM) for hybrid renewable energy site selection in Tanzania.

SKILLS

Research Design

Literature review, data collection via surveying and web scraping, data engineering, statistical modeling and analysis, academic writing.

Quantitative Analysis

Statistical modeling with OLS, GLS, GLM, GLMM; spatial-temporal modeling with Bayesian/INLA; time-series and network analysis (QAP, ERGM); causal inference (DID, SCA, RDD, PSM, 2SLS); dimensionality reduction (PCA, Factor Analysis, Ridge, Lasso); robustness testing (permutation testing, Monte Carlo simulation, bootstrapping).

Machine Learning & AI

Agentic AI with Cursor, LangGraph, Claude Code; deep learning (CNNs) for image classification; transformer-based NLP (RoBERTa) for text and sentiment analysis; gradient boosting (XGBoost); LLM/GPT API pipelines for data extraction, distillation, and similarity checking; supervised/unsupervised methods (KNN, Random Forest, SVM, Bagging, K-Means); text classification, topic modeling, content analysis.

Geospatial Analysis

Remote sensing, radiometric and geometric correction, image enhancement, thematic information extraction, spatial object classification, change detection, WebGIS, suitability modeling, multi-criteria decision making (MCDM), surface analysis, spatial and areal interpolation, geocoding (Nominatim/OpenStreetMap).

Programming & Tools

R, Python, Stata, PHP, Java, C/C++/C#; SPSS, Tableau, Power BI; ArcGIS Pro, QGIS, Google Earth Engine, YOLO/RoboFlow, Drone2Map, Google OR-Tools; database design on MySQL, PostgreSQL, SQLite.

EDUCATION

PhD, Political Science, University of Texas at Dallas (expected Fall 2026)

MS, Geospatial Information Science, University of Texas at Dallas

MS, Social Data Analytics and Research, University of Texas at Dallas

MBA, Strategic Marketing, California State University

MS, Computer Engineering, Ohio State University

MS, Thermodynamics, Shanghai Jiao Tong University

BS, Marine Power Engineering, Shanghai Jiao Tong University

ACHIEVEMENTS

Fred Hill Endowed Scholarship, 2025

Certificate in Graduate Teaching, 2024